Evaluation runs: zab (zab-broadcast-remix)

generated 2026-11-03 from brain-data 03b0b57cc 2026-11-03 11:21:51 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): . Blue rows are at the current pins (TraceForge 9a0629e, merge join 367cdc6, JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	2	161	conforms	0.0	2	1 h, 250 GB	2	3	brain03, 2026-11-02
2	6	33161	conforms	0.9	4	1 h, 250 GB	6	21	brain03, 2026-11-02
3	20	9550181	conforms	149.9	52	1 h, 250 GB	20	210	brain03, 2026-11-02
4			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **issue 1**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	2	128	10	violation	0.0	2	1 h, 250 GB	12	23	brain03, 2026-11-02
2	6	24215	70	violation	0.6	2	1 h, 250 GB	76	441	brain03, 2026-11-02
3	20	6881334	258	violation	105.8	36	1 h, 250 GB	278	5370	brain03, 2026-11-02
4				🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 correct fifo, 4 issue1-propose-to-ackld fifo, 4 fifo issue1-propose-to-ackld, 5 fifo correct, 5 fifo issue1-propose-to-ackld, 6 fifo correct, 6 fifo issue1-propose-to-ackld, 7 fifo correct, 7 fifo issue1-propose-to-ackld.

skipped: a smaller size had reached its limit: 5 correct fifo, 5 issue1-propose-to-ackld fifo, 6 correct fifo, 6 issue1-propose-to-ackld fifo, 7 correct fifo, 7 issue1-propose-to-ackld fifo, 8 correct fifo, 8 issue1-propose-to-ackld fifo.

stateful

stateful: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	2	161	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2	6	33161	conforms	1.0	4	1 h, 250 GB	brain02, 2026-11-02
3	20	9550181	conforms	191.0	48	48 h, 250 GB	brain05, 2026-11-02
4			🕓 (1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **issue 1**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1	2	128	10	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2	6	24215	70	violation	0.7	2	1 h, 250 GB	brain02, 2026-11-02
3	20	6881334	258	violation	121.7	41	48 h, 250 GB	brain05, 2026-11-02
4				(1 h)			1 h, 250 GB	brain02, 2026-11-02

Not run (stateful). left out by the table rules (ok): 3 correct fifo, 3 issue1-propose-to-ackld fifo.

skipped: a smaller size had reached its limit: 5 correct fifo, 5 issue1-propose-to-ackld fifo, 6 correct fifo, 6 issue1-propose-to-ackld fifo, 7 correct fifo, 7 issue1-propose-to-ackld fifo, 8 correct fifo, 8 issue1-propose-to-ackld fifo.

planned but never scheduled (grid points beyond the ladder): 4 fifo correct, 4 fifo issue1-propose-to-ackld, 5 fifo correct, 5 fifo issue1-propose-to-ackld, 6 fifo correct, 6 fifo issue1-propose-to-ackld, 7 fifo correct, 7 fifo issue1-propose-to-ackld.

offline

offline: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	161	conforms	0.3	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
2	6	33161	conforms	1.4	4	1 h, 250 GB	0.1	1.3	0.0	merge	brain05, 2026-11-03
3	20	9550181	conforms	192.9	50	1 h, 250 GB	0.1	192.8	0.0	merge	brain05, 2026-11-03

offline: pair **issue 1**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	128	2	violation	0.3	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
2	6	24215	5	violation	0.9	4	1 h, 250 GB	0.1	0.8	0.0	merge	brain05, 2026-11-03
3	20	6881334	9	violation	122.0	40	1 h, 250 GB	0.1	121.8	0.0	merge	brain05, 2026-11-03

Not run (offline). skipped: a smaller size had reached its limit: 4 correct fifo, 4 issue1-propose-to-ackld fifo, 5 correct fifo, 5 issue1-propose-to-ackld fifo, 6 correct fifo, 6 issue1-propose-to-ackld fifo, 7 correct fifo, 7 issue1-propose-to-ackld fifo, 8 correct fifo, 8 issue1-propose-to-ackld fifo.

planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 fifo issue1-propose-to-ackld, 5 fifo correct, 5 fifo issue1-propose-to-ackld.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1		161	conforms	0.1	4	10 min, 250 GB		0.1	0.1	in-memory	brain01, 2026-11-03
2		33161	conforms	0.1	4	10 min, 250 GB		1.3	0.1	in-memory	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **issue 1**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1		128	2	violation	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
2		24215	5	violation	0.0	4	10 min, 250 GB		0.8	0.0	in-memory	brain01, 2026-11-03

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 3 fifo correct.jmcspec-tfimpl, 3 fifo issue1-propose-to-ackld.jmcspec-tfimpl, 4 fifo correct.jmcspec-tfimpl, 4 fifo issue1-propose-to-ackld.jmcspec-tfimpl, 5 fifo correct.jmcspec-tfimpl, 5 fifo issue1-propose-to-ackld.jmcspec-tfimpl.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	161	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
2	6	33161	conforms	1.1	4	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **issue 1**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	2	128	10	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
2	6	24215	70	violation	0.7	4	1 h, 250 GB	2	brain01, 2026-11-02

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 fifo issue1-propose-to-ackld, 5 fifo correct, 5 fifo issue1-propose-to-ackld.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1		161	conforms	0.1	4	10 min, 250 GB		0.1	0.1	in-memory	brain01, 2026-11-03
2		33161	conforms	0.0	4	10 min, 250 GB		1.3	0.0	in-memory	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **issue 1** against **issue1-propose-to-ackld**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1		128	conforms	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
2		24215	conforms	0.0	4	10 min, 250 GB		0.8	0.0	in-memory	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **issue 1**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1		128	2	violation	0.2	4	10 min, 250 GB		0.1	0.2	in-memory	brain01, 2026-11-03
2		24215	5	violation	0.0	4	10 min, 250 GB		0.8	0.0	in-memory	brain01, 2026-11-03

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 3 fifo correct.jimpl-tfimpl, 3 fifo issue1-propose-to-ackld.jimpl-tfimpl, 3 fifo issue1-propose-to-ackld.vs-correct.jimpl-tfimpl, 4 fifo correct.jimpl-tfimpl, 4 fifo issue1-propose-to-ackld.vs-correct.jimpl-tfimpl, 4 fifo issue1-propose-to-ackld.jimpl-tfimpl.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	161		conforms	0.0	4	10 min, 250 GB	0.1		0.0	in-memory	brain01, 2026-11-03
2	33161		conforms	0.0	4	10 min, 250 GB	1.3		0.0	in-memory	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **issue 1** against **issue1-propose-to-ackld**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	128		conforms	0.0	4	10 min, 250 GB	0.1		0.0	in-memory	brain01, 2026-11-03
2	24215		conforms	0.5	4	10 min, 250 GB	0.8		0.5	in-memory	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **issue 1**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	161		2	violation	0.1	4	10 min, 250 GB	0.1		0.1	in-memory	brain01, 2026-11-03
2	33161		5	violation	0.0	4	10 min, 250 GB	1.3		0.0	in-memory	brain01, 2026-11-03

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 3 fifo correct.tfimpl-jimpl, 3 fifo issue1-propose-to-ackld.vs-correct.tfimpl-jimpl, 3 fifo issue1-propose-to-ackld.tfimpl-jimpl, 4 fifo correct.tfimpl-jimpl, 4 fifo issue1-propose-to-ackld.tfimpl-jimpl, 4 fifo issue1-propose-to-ackld.vs-correct.tfimpl-jimpl.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1		161	conforms	0.1	4	10 min, 250 GB		0.1	0.1	in-memory	brain01, 2026-11-03
2		33161	conforms	0.0	4	10 min, 250 GB		1.3	0.0	in-memory	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **issue 1** against **issue1-propose-to-ackld**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1		128	conforms	2.9	4	10 min, 250 GB		0.1	2.9	in-memory	brain01, 2026-11-03
2		24215	conforms	0.2	4	10 min, 250 GB		0.8	0.2	in-memory	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **issue 1**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1		128	2	violation	0.1	4	10 min, 250 GB		0.1	0.1	in-memory	brain01, 2026-11-03
2		24215	5	violation	0.0	4	10 min, 250 GB		0.8	0.0	in-memory	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): 3 fifo correct.jimpl-tfimpl, 3 fifo issue1-propose-to-ackld.jimpl-tfimpl, 3 fifo issue1-propose-to-ackld.vs-correct.jimpl-tfimpl, 4 fifo correct.jimpl-tfimpl, 4 fifo issue1-propose-to-ackld.vs-correct.jimpl-tfimpl, 4 fifo issue1-propose-to-ackld.jimpl-tfimpl.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	161		conforms	0.0	4	10 min, 250 GB	0.1		0.0	in-memory	brain01, 2026-11-03
2	33161		conforms	0.1	4	10 min, 250 GB	1.3		0.1	in-memory	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **issue 1** against **issue1-propose-to-ackld**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	128		conforms	0.0	4	10 min, 250 GB	0.1		0.0	in-memory	brain01, 2026-11-03
2	24215		conforms	0.0	4	10 min, 250 GB	0.8		0.0	in-memory	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **issue 1**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	161		2	violation	0.1	4	10 min, 250 GB	0.1		0.1	in-memory	brain01, 2026-11-03
2	33161		5	violation	0.2	4	10 min, 250 GB	1.3		0.2	in-memory	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). planned but never scheduled (grid points beyond the ladder): 3 fifo correct.tfimpl-jimpl, 3 fifo issue1-propose-to-ackld.vs-correct.tfimpl-jimpl, 3 fifo issue1-propose-to-ackld.tfimpl-jimpl, 4 fifo correct.tfimpl-jimpl, 4 fifo issue1-propose-to-ackld.tfimpl-jimpl, 4 fifo issue1-propose-to-ackld.vs-correct.tfimpl-jimpl.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	161	161	conforms	9.1	159	1 h, 250 GB		brain01, 2026-11-03
2	33161	33161	conforms	211.0	16339	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **issue 1** against **issue1-propose-to-ackld**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	128	128	conforms	8.6	157	1 h, 250 GB		brain01, 2026-11-03
2	24215	24215	conforms	184.1	11590	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **issue 1**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	161	128	10	violation	8.3	173	1 h, 250 GB	105	brain01, 2026-11-03
2	33161	24215	70	violation	190.4	11039	1 h, 250 GB	227	brain01, 2026-11-03

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). planned but never scheduled (grid points beyond the ladder): 3 fifo issue1-propose-to-ackld.w16, 3 fifo issue1-propose-to-ackld.vs-correct.w16, 3 fifo correct.w16, 4 fifo issue1-propose-to-ackld.w16, 4 fifo issue1-propose-to-ackld.vs-correct.w16, 4 fifo correct.w16.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	161	161	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02
2	33161	33161	conforms	1.1	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **issue 1** against **issue1-propose-to-ackld**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	128	128	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
2	24215	24215	conforms	0.7	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **issue 1**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	161	128	10	violation	0.0	4	1 h, 250 GB	1	brain01, 2026-11-02
2	33161	24215	70	violation	0.7	6	1 h, 250 GB	2	brain01, 2026-11-02

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export). planned but never scheduled (grid points beyond the ladder): 3 fifo correct, 3 fifo issue1-propose-to-ackld, 4 fifo correct, 4 fifo issue1-propose-to-ackld.